

Loupe — Product & Business Requirements

A digital museum of “how things actually work”

*Working title: **Loupe** (a jeweler’s loupe — a small lens you hold up to look closer at something.*

Placeholder — see naming options at the end.)

1. Executive Summary

Loupe is a website made of small, self-contained, interactive “exhibits” — each one taking a single everyday mechanism (what happens when you click a button, tap a card, send a text, cross a border on a flight) and walking a visitor through it, layer by layer, in 5–10 minutes. It is not a course platform. There are no skill trees, no lessons that build on each other, no accounts required to start. You arrive curious about one specific thing, you leave understanding that one thing, and you can leave whenever you want. Think of a museum wing, not a syllabus.

This document covers the product concept, how a single exhibit should be structured, the competitive landscape, the tech stack, a phased build plan sized for a solo builder with a full-time job, and a monetization strategy — grounded in research on comparable products and the microlearning market as of mid-2026.

2. Why This, Why Now

The instinct behind Loupe — that people will happily spend a few focused minutes learning something with no test at the end, if it’s presented well — is already validated by several independent, unrelated products:

- **Neal.fun**, a solo-built site of small interactive experiments, reportedly draws over 4 million visits a month with zero paywalls, funded entirely by ads and sponsorships — proof that “one page, one idea, no login” travels well organically.
- **Ciechanow.ski**, Bartosz Ciechanowski’s blog of hand-built interactive articles on engineering and physics topics (internal combustion engines, cameras, gears), is sustained by reader patronage on Patreon — proof that deep, well-crafted technical explainers earn direct support from a niche, technical audience.
- **Nicky Case and the “Explorable Explanations” movement** turned this into a small open-source ecosystem (tools like Loopy and Idyll, a community jam, a curated directory at explorabl.es) — proof there’s an entire design language here, not just isolated hits.
- **Brilliant.org** has scaled interactive, problem-first STEM learning into a subscription business

(roughly \$25–28/month, reportedly 10M+ users) — proof people pay recurring money for hands-on learning that isn't video lectures.

- **Nibble**, a newer app, is the closest direct competitor: 5–10 minute interactive lessons across 20+ broad topics, gamified, subscription around \$12/month. It validates the exact session length and format Loupe is aiming for — but leans toward breadth, quizzes, and streak-driven habit mechanics rather than depth and craft.

On the market side, independent research firms put the global microlearning market anywhere from roughly \$2–6 billion in 2026 depending on methodology, growing at a 12–22% CAGR — a wide range because most of that number is enterprise L&D and compliance training, not consumer curiosity products. That's useful context, not a number to lean on for a pitch deck: it tells you the *format* (short, interactive, mobile) is proven at scale, even though most of the dollars are corporate, not the audience Loupe is going after.

The gap Loupe can occupy: everything above is either (a) free but ad-funded and unpredictable as a business, (b) a broad, gamified habit-loop app, or (c) a solo passion project with no ambition to be a product. Nobody has combined *museum-grade craft* (Ciechanowski's depth) with a *real content system and business model* (Brilliant/Nibble's discipline) at *normal-person, no-prerequisite scope* (Neal.fun's accessibility).

3. Competitive Landscape

Product	Format	Session length	Monetization	What to borrow	What to avoid
Neal.fun	Single-page interactive toys	1–5 min	Ads/sponsorship	No login, instant payoff, shareability	No sustainable content system, revenue ceiling is low
Ciechanow.ski	Long-form interactive articles	15–40 min	Patreon	Craftsmanship, trust, custom visuals per topic	Too slow to produce for a content <i>business</i> ; sessions run longer than your target
Nicky Case / explorabl.es	Open-source explorable explanations	Varies	Patreon, mostly non-commercial	The design vocabulary (sliders, direct manipulation, "ladder of	Not built as a product — no discovery, no account system, no

				abstraction")	roadmap
Brilliant.org	Structured STEM courses	Ongoing, course-based	Subscription (~\$25–28/mo)	Rigorous interactivity, proof people pay	STEM-only, course structure is the opposite of “one-way”
Nibble	Broad topic microlearning app	5–10 min	Subscription (~\$12/mo)	Session length, format mix (text/audio/video/quiz)	Breadth-over-depth, gamification-first, quiz-heavy rather than genuinely explanatory
Corporate training tools (Articulate, Elucidat, Captivate)	Authoring tools for interactive slides	Varies	B2B licenses	The “flow that keeps a story going” the user referenced	Built for compliance, not curiosity — flat, corporate visual language

Loupe’s wedge: depth and craft like Ciechanowski, session length and mobile-first packaging like Nibble, shareability and no-login-required like Neal.fun — and, critically, a content moat competitors can’t easily copy: exhibits written from *real, first-hand expertise* rather than generic research. More on this below.

4. The Unfair Advantage: Start With What You Already Know

Given your background — 3.5+ years as a backend engineer, prior work leading fraud & risk, SSO, and loyalty systems at Tata Digital, plus a deep learning research background — the strongest opening wing isn’t random trivia. It’s **systems and machines**: the button-click exhibit is the template, not a one-off. You can write, with real authority, exhibits like:

- *What happens when you tap your card at a store* (authorization, network routing, fraud scoring, settlement — this is literally your former domain)
- *What happens when you log in with “Sign in with Google”* (SSO/OAuth handshake — also your domain)
- *What happens when your phone flags a transaction as fraud*
- *What happens when you send a UPI payment*

- *What happens when a website checks if you're a bot*

This matters commercially, not just creatively: AI-generated content mills can produce generic “how does WiFi work” copy all day. They can’t produce a fraud-detection exhibit written by someone who actually built fraud-detection systems. That authenticity is your defensibility in a world where content is about to get much cheaper to fake.

5. Target Users

Primary: “The Idle Curious.” Someone scrolling social media on a commute or a break, who clicks something because the headline made them go “wait, how *does* that work?” They are not trying to learn a skill. Success = they finish, feel smarter, and share it or come back tomorrow for a different one.

Secondary: “The Technical Explainer-seeker.” Developers, engineers, and technical hobbyists specifically looking for well-made explanations of systems concepts (the audience Ciechanowski and sites like your own professional network already trust you with). This group is smaller but higher-intent, more likely to pay, and more likely to share within technical communities (Hacker News, r/programming, dev Twitter/X).

Tertiary, later: L&D and education buyers. Once a library of well-made exhibits exists, corporate onboarding teams and educators are a plausible licensing audience — see Section 12.

6. Core Concept: The Exhibit

6.1 Anatomy of an exhibit — the button-click example, fully broken down

This is the flagship worked example and should be exhibit #1:

1. **Hook (10–15 sec):** “You just clicked a button. In the next few minutes, you’ll follow that one click all the way down to electricity moving through silicon — and back up again.”
2. **UI/DOM layer:** the click fires a browser event; a listener callback gets queued.
3. **JavaScript engine layer:** the callback is parsed/compiled and executed; the event loop hands off control.
4. **Operating system layer:** the browser process makes system calls; the OS scheduler allocates CPU time and memory.
5. **Hardware layer:** instructions become machine code; logic gates (AND/OR/NOT combined into an ALU) switch transistor states; electrical pulses actually move.
6. **Back up the stack:** a result triggers a repaint, the GPU rasterizes pixels, the screen updates — the full round trip happens in milliseconds.

7. **Takeaway (15–20 sec):** one memorable line that survives after the visitor closes the tab — e.g., “every click is a tiny, controlled lightning storm, happening billions of times a second.”

Notice the shape: this isn’t a story that could branch — it’s a **ladder**, one layer under the next. That’s not incidental. It’s the reusable interaction pattern described next.

6.2 Reusable interaction templates (your production toolkit)

Building bespoke UI for every exhibit doesn’t scale for a solo builder. Instead, define a small set of interaction “shapes,” each a reusable component that different exhibit content can be poured into:

Template	Shape	Good for	Example
The Zoom Ladder	Scroll drills from macro → micro (or vice versa)	Systems with clean layers	Button click, card tap, DNS lookup
The Timeline Scroll	Chronological unfold	Processes with a clear sequence over time	A package’s journey, a transaction settling
The Compare Toggle	Two states, flip a switch	Concepts best understood by contrast	TCP vs UDP, symmetric vs asymmetric encryption
The Tweak & Watch	User adjusts a variable, sees the outcome change live	Concepts with a cause-effect relationship	Compound interest, network latency, load balancing
The Anatomy Cutaway	A static object with clickable hotspots	Physical or structural objects	Inside a credit card chip, anatomy of a URL

Five templates, well-built once, can host dozens of exhibits — this is the single biggest lever for making a solo, part-time production pace realistic.

6.3 The 5–10 minute session template

Every exhibit follows the same narrative beats regardless of which interaction template it uses:

1. Hook (~15 sec) — a question or a surprising claim
2. Context (~30–45 sec) — the bare minimum background needed, no prerequisites assumed
3. The walkthrough (~4–6 min) — 3–7 interactive steps/layers, the core of the exhibit
4. The takeaway (~15–20 sec) — one line worth remembering
5. Optional “go deeper” (opt-in only, never required) — links to a source article or a related exhibit
6. “You might also like...” — 2–3 related exhibits, to chain sessions (this is where return visits come

from)

6.4 Topic selection rubric — what qualifies as an exhibit

A candidate topic should pass most of these:

- **Ubiquity:** something people encounter regularly but have never seen the inside of
- **Mechanism, not opinion:** a factual process, not a debate — this keeps the format out of contested territory entirely
- **Layerable:** decomposes cleanly into 3–7 sequential or hierarchical steps
- **Terminal, not a skill:** after 8 minutes the visitor “gets it” — no follow-up course implied. This is the core difference from Duolingo/Brilliant’s skill-tree model, and it’s also why “one-way” is the right word for these: they don’t loop the visitor back for more instruction.
- **Visualizable:** has a spatial, physical, or flow-based structure that can actually be drawn or animated well
- **Honestly boundable:** can be told in under 10 minutes without becoming misleading through oversimplification

Topics that fail this rubric (abstract philosophy, contested history, anything needing nuance a 7-step ladder can’t hold) should be rejected or saved for a different, longer-form template later — don’t force everything into the museum format.

7. Information Architecture & UX

- **Home = the museum lobby.** Wings (topic categories) shown as visual entry points, not a search bar first. First wing: “Systems & Machines.”
- **No account required to view any exhibit.** Frictionless entry is the single most important UX decision — it’s what made Neal.fun and Wikipedia’s “random article” work.
- **Every exhibit has a clean, shareable URL.** No routing through the homepage.
- **Progress indicator** inside each exhibit (step counter or scroll progress), so the 5–10 minute promise feels honest and finishable.
- **“Surprise me” button** — a random-exhibit mechanic, borrowed deliberately from Wikipedia/Neal.fun, for pure top-of-funnel virality.
- **Lightweight accounts (magic link, no password)** only for saving a personal “collection” of exhibits — introduced in Phase 2, not Phase 1.
- **Related exhibits** at the end of every session to chain visits.
- **Text-only fallback** for every visual sequence — both an accessibility requirement and a low-

bandwidth requirement, relevant given a lot of the mobile-first audience this format targets will be on inconsistent connections.

8. Functional & Non-Functional Requirements

Must-have (MVP):

- FR1: Static, SEO-indexable exhibit pages, server-rendered
- FR2: At least one working interaction template (Zoom Ladder) as a reusable component, not hardcoded per exhibit
- FR3: Mobile-first touch interactions (tap, swipe, drag — not hover-dependent)
- FR4: A “content schema” so new exhibits are authored as structured data (title, layers, narration, interaction type) rather than hand-coded pages
- FR5: Basic analytics: completion rate and drop-off point per exhibit step

Should-have (Phase 2):

- FR6: Lightweight accounts + saved collections
- FR7: Search/filter by wing, topic, duration
- FR8: “Surprise me” random exhibit

Non-functional:

- NFR1: First interaction visible in under ~2 seconds on a mid-range phone on 4G — aggressive lazy-loading of any 3D/WebGL assets, with a lighter-weight fallback for low-end devices
- NFR2: Accessibility — alt text/captions for every visual sequence, keyboard navigation, a reduced-motion mode
- NFR3: Each exhibit is effectively **append-only** once published — no obligation to maintain or update old exhibits indefinitely. This mirrors the “one-way” framing you used and is a deliberate sustainability choice for a solo content producer (Nicky Case has written publicly about the burden of feeling obligated to keep updating old explorables forever — designing this out from day one avoids that trap).

9. Technical Architecture

Given the 3D/scrollytelling ambitions here and the overlap with the immersive 3D portfolio site you’re already planning, the stacks should deliberately share skills and possibly components:

- **Frontend:** Next.js (React + TypeScript) for server rendering and SEO

- **3D layer:** React Three Fiber + drei, used selectively (Tier 2/3 exhibits only — see below), reusing whatever you build for the portfolio's interactive 3D research showcase
- **Scroll/motion:** GSAP ScrollTrigger (current stable line handles mobile pinning much better than older versions) + Lenis for smooth scroll; CSS `view-timeline` for simpler scroll effects to keep JS bundle size down where full GSAP isn't needed
- **Content model:** exhibits authored as structured MDX/JSON against the "exhibit schema" from FR4 — this decouples content from the interaction template code, so the Zoom Ladder component can render *any* exhibit's layer data without new engineering per exhibit
- **Hosting:** static-first (Vercel/Cloudflare Pages) with a CDN for compressed 3D assets (Draco/Meshopt-compressed glTF) — durable, cheap at solo-founder scale
- **Backend:** minimal — auth (magic link), saved collections, and an analytics event store; no need for anything heavier at this stage

Not every exhibit needs 3D. Define production tiers up front so scope stays sane:

- **Tier 0:** text + static illustration + scroll-triggered CSS reveal (cheapest, fastest to produce)
- **Tier 1:** 2D animated/interactive (SVG/Canvas, draggable sliders)
- **Tier 2:** lightweight 3D (React Three Fiber, simple geometry)
- **Tier 3:** full cinematic 3D/WebGL scrollytelling (reserved for flagship exhibits only — this is where the button-click piece belongs)

Content pipeline synergy: this is a genuinely good first real use case for the multi-agent AI system you're already planning — a research agent and fact-check agent can draft the raw layer-by-layer breakdown and verify technical claims for a new exhibit, which you then turn into the interactive script and visual design. That directly exercises the "one prompt → researched, fact-checked output" loop you wanted from that system, on a concrete product instead of an abstract one.

10. Phased Roadmap (sized for a solo builder with a full-time job)

Phase 0 — Prove the format (2–4 weeks): Build exhibit #1 (button click) by hand as a single static page. No accounts, no CMS, no schema yet — just prove the Zoom Ladder pattern works and feels good on mobile. Ship it to Show HN / r/webdev / relevant dev communities and watch completion rate and drop-off point. This is the single highest-signal step in the whole plan — don't build anything else until this exhibit's data says the format works.

Phase 1 — MVP library (2–3 months): 8–12 exhibits across 1–2 wings (start with Systems & Machines, your unfair advantage). Build the content schema and at least 2 reusable templates. No monetization yet — the goal is an audience, an email list, and SEO foundation.

Phase 2 (next 3–6 months): Accounts, saved collections, a paid membership tier (see below), 20–30 exhibits, and a public "suggest a topic" mechanism to crowdsource ideas.

Phase 3 (6–12+ months): B2B angle — licensing or embedding exhibits into engineering onboarding docs or corporate L&D, where your domain credibility (fraud/risk, SSO, payments) is a real sales advantage, not just content flavor.

11. Business Model & Monetization

None of the researched comparables monetize the same way, which is useful — it means there's a real choice to make rather than one obvious playbook:

Model	Precedent	Pros	Cons
Ads/sponsorship	Neal.fun	No friction for visitors, scales with traffic	Needs millions of monthly visits to matter; unpredictable
Patronage (Patreon-style)	Ciechanowski, Nicky Case	Keeps everything free/open, aligns with "museum" identity	Revenue ceiling tied to fan generosity, not built for scale
Subscription/membership	Brilliant (~\$25–28/mo), Nibble (~\$12/mo)	Proven willingness to pay for this exact format	Conflicts somewhat with "free general admission" museum framing if applied too broadly
B2B licensing	Corporate training tools, microlearning platforms (mostly enterprise-funded)	Largest addressable dollars (most of the microlearning market is enterprise)	Longest to build credibility and sales motion for

Recommendation — the real museum model: most museums have free general admission *and* a paid membership for perks, *and* they license traveling exhibits to other institutions. Map that directly:

1. **All exhibits free forever.** This is core to the identity, not just a growth tactic — it's what makes "digital museum" true rather than a marketing line.
2. **Optional membership** (a "Collector's Pass") for perks that don't gate core content: offline packs, early access to new exhibits, ad-free browsing (if ads are ever introduced), and the ability to request/vote on topics. This is where Nibble/Brilliant pricing gives you a real anchor — something in the \$3–8/month range fits a "supporter" tier better than trying to match Brilliant's \$25+/month course-platform pricing, especially with regional/PPP-adjusted pricing for markets like India where a flat US-dollar price underperforms.
3. **B2B/licensing later** as the real revenue engine once the library and reputation exist — this is

where the bulk of the actual microlearning market's dollars sit, and it's the piece your professional background gives you a genuine edge in selling.

12. Market Sizing (directional, not a forecast)

- **TAM:** global microlearning market, roughly \$2–6B in 2026 across independent estimates, growing 12–22% CAGR — but this figure is overwhelmingly enterprise L&D and compliance training, not consumer curiosity content.
 - **SAM:** the consumer edutainment/curiosity subscription slice — hard to size precisely, but Brilliant's reported 10M+ user base at STEM-only scope, and Nibble's emergence as a broader competitor, suggest a real market in the low-hundreds-of-millions-of-dollars range for consumer bite-sized-learning subscriptions specifically.
 - **SOM (realistic, bootstrapped, year 1–2):** a few thousand to low tens of thousands of monthly visitors from organic/community sharing in year one if Phase 0–1 land well; a few hundred to low thousands of paying members by year two if membership is introduced on schedule. These are deliberately modest, solo-builder-scale numbers — not a pitch-deck projection.
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13. Metrics That Matter

North Star: exhibits completed per returning visitor per week (measures both quality — do people finish? — and habit — do they come back?).

Supporting metrics:

- Completion rate per exhibit (target >60%; anything lower is a signal that exhibit's pacing or interaction is off)
 - Drop-off point within an exhibit (which layer/step loses people — the most useful content-quality signal you'll have)
 - Median session length (should land in the 5–10 min target band)
 - Share rate (socials, community links)
 - 7-day and 30-day return rate
 - Email/collection capture rate
 - Later: membership conversion rate, qualitative feedback/NPS
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14. Risks & Open Questions

- **Production bottleneck:** high-fidelity exhibits are slow to build solo. Mitigate with the tiered

production system (Section 9) and AI-assisted first drafts — don't let every exhibit aim for Tier 3.

- **Trust risk:** getting a technical fact wrong on a niche topic erodes credibility fast, especially with the technical audience this format is likely to attract first. Build a fact-check pass into the pipeline (your multi-agent system's fact-check agent is directly useful here) and consider a visible "sources" footer per exhibit.
 - **Discovery risk:** no algorithmic feed means growth depends on shareability, SEO, and community posting (Show HN, Product Hunt, relevant subreddits) — budget real time for this, it won't happen passively.
 - **Scope creep risk:** the temptation to add streaks, gamification, and full accounts early (like Nibble) works against the "museum, not habit app" identity — resist until Phase 2 at the earliest, and only add what serves genuine return visits rather than dark-pattern engagement.
 - **Monetization ceiling:** a free-first identity caps how aggressively membership can be priced or gated — B2B is where the real business case has to eventually live.
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15. Naming Options

A few directions, all placeholders until you land on one:

- **Loupe** — a jeweler's tool for looking closer; short, brandable, literal
 - **Substrate** — "the layer underneath," fits the systems/mechanism angle well
 - **First Principles** — descriptive, slightly more serious/academic tone
 - **The Undercurrent** — evokes "what's really flowing beneath the surface"
 - **Aperture** — photography metaphor for "looking closer," slightly overused in tech branding
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16. Immediate Next Steps

1. Write the full layer-by-layer script for the button-click exhibit as plain text first — no code, no design — to validate the narrative before building anything.
2. Prototype the Zoom Ladder interaction template in React Three Fiber, reusing whatever setup you build first for the 3D portfolio site.
3. Ship exhibit #1 as a single static page on a simple domain/subdomain — no accounts, no CMS — and share it in 2–3 relevant developer communities.
4. Only after real completion-rate and drop-off data comes back from exhibit #1, decide whether to invest in the full content schema and second interaction template.

